Ratios in Algebra

Fluency

Simplify

Worked example: Simplify $12xy : 18y$. Divide both sides by the HCF, $6y$: $12xy : 18y = 2x : 3$.

(a) $6x : 8x$

(b) $10x : 15x$

(c) $4x : 12x$

(d) $9x : 6x$

(e) $5xy : 10y$

(f) $8x^2 : 4x$

(g) $3x : 9x^2$

(h) $12ab : 18b$

Find the unknown

Worked example: If $x : y = 2 : 5$ and $x = 8$, find y. Since $8 \div 2 = 4$, multiply the other side by 4 too: $y = 5 \times 4 = 20$.

(a) $x : y = 3 : 4$. If $x = 15$, find y.

(b) $a : b = 5 : 2$. If $b = 8$, find a.

(c) $p : q = 2 : 7$. If $q = 21$, find p.

(d) $m : n = 9 : 4$. If $m = 27$, find n.

(e) $x : y = 7 : 3$. If $y = 18$, find x.

(f) $a : b = 11 : 5$. If $a = 55$, find b.

Problem Solving

Solve

Worked example: Solve $(x+2) : (x-1) = 5 : 3$. Cross-multiply, then solve the equation:

$$\begin{aligned} 3(x+2) &= 5(x-1) \\ 3x+6 &= 5x-5 \\ 11 &= 2x \\ x &= 5.5. \end{aligned}$$

(a) $(x+3) : (x-1) = 3 : 2$

(b) $(3x+1) : (x+7) = 2 : 1$

Sharing in a ratio

Worked example: Share \$80 in the ratio $3 : 5$. The parts are $3k$ and $5k$, so $3k + 5k = 80 \Rightarrow k = 10$. The shares are $3 \times 10 = \$30$ and $5 \times 10 = \$50$.

(a) A 40 cm line segment is divided in the ratio $3 : 5$. Find the length of each part.

(b) Three siblings share \$780 in the ratio $2 : 3 : 8$. How much does each receive?

Word problems

Worked example: $a : b = 5 : 3$ and $a - b = 14$. Find a and b. Write $a = 5k$ and $b = 3k$, so $5k - 3k = 14 \Rightarrow k = 7$. Then $a = 35$ and $b = 21$.

(a) The ratio of cats to dogs at a shelter is $3 : 4$. There are 12 more dogs than cats. How many of each are there?

(b) Henry and Jack share some money in the ratio $3 : 5$. Henry receives \$45. How much does Jack receive, and what is the total amount shared?

Reasoning

No algebra here — just ratio reasoning. Show your working.

(a) Explain why the ratios $8 : 12$ and $2 : 3$ represent the same relationship.

(b) Cordial A is mixed 1 part cordial to 4 parts water. Cordial B is mixed 2 parts cordial to 9 parts water. Which is the stronger-tasting cordial? Justify your answer using equivalent fractions or ratios.

(c) A fruit drink is made from orange and mango juice in the ratio $5 : 3$. Another is made in the ratio $7 : 4$. Which drink has the higher proportion of orange juice? Justify your answer.

(d) Paint colour "Sunrise" is mixed from red and yellow paint in the ratio $3 : 7$. You want to make a larger batch using 18 L of red paint. How much yellow paint do you need?

(e) A map has a scale of $1 : 25\,000$. Explain what this ratio means, then calculate the real-world distance, in km, represented by 6 cm on the map.

Enrichment

Combining Ratios

To combine $a : b$ and $b : c$ into one ratio $a : b : c$, scale each ratio so that the value of b matches in both.

Worked example: $a : b = 2 : 3$ and $b : c = 4 : 5$. The lowest common multiple of 3 and 4 is 12, so scale the first ratio by 4 and the second by 3: $a : b = 8 : 12$ and $b : c = 12 : 15$. Combining these, $a : b : c = 8 : 12 : 15$.

(a) $a : b = 3 : 4$ and $b : c = 2 : 5$. Find $a : b : c$.

(b) $x : y : z = 2 : 5 : 9$. If $z - x = 42$, find x, y and z.

A Harder Ratio Problem

Worked example: Three years ago, the ratio of Mia's age to her brother's age was $5 : 2$. In two years' time, the ratio will be $2 : 1$. Find their current ages. Let Mia's current age be M and her brother's be B.

$$\text{3 years ago: } \frac{M-3}{B-3} = \frac{5}{2} \;\Rightarrow\; 2(M-3) = 5(B-3) \;\Rightarrow\; M = \frac{5B-9}{2}.$$

$$\text{In 2 years: } \frac{M+2}{B+2} = \frac{2}{1} \;\Rightarrow\; M+2 = 2(B+2) \;\Rightarrow\; M = 2B+2.$$

Setting these equal: $\dfrac{5B-9}{2} = 2B+2 \Rightarrow 5B-9 = 4B+4 \Rightarrow B = 13$, so $M = 2(13)+2 = 28$.

(a) Six years ago, the ratio of Ana's age to Ben's age was $3 : 1$. In four years' time, the ratio will be $2 : 1$. Find their current ages.
